Code No: 117BX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
COMPUTER GRAPHICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Outline the concepts of Points and lines.	[2]
b)	Describe the working of graphics monitors and workstations in brief	[3]
c)	What is viewing coordinate reference frame?	[2]
d)	Discuss the matrix representations and homogeneous coordinates.	[3]
e)	Summarize the sweep representations.	[2]
f)	Discuss the composite transformations.	[3]
g)	Outline the concept of Clipping.	[2]
h)	List the polygon rendering methods.	[3]
i)	List the pros and cons of animation.	[2]
j)	List various computer animation languages.	[3]

PART – B

(50 Marks)

2.a) Calculate the points between the starting coordinates (20, 10) and ending coordinates (30, 18) by using Bresenham's Line Algorithm.
b) Differentiate between the raster-scan systems and random scan systems. [5+5]

OR

3.a) Illustrate the working of Cathode-Ray Tubes (CRT).
b) Write the Flood-Fill Algorithm with an example. [5+5]

4.a) Given a circle C with radius 10 and center coordinates (1, 4). Apply the translation with distance 5 towards X axis and 1 towards Y axis. Obtain the new coordinates of C without changing its radius.
b) Discuss the window to view-port coordinate transformation. [5+5]

OR

5.a) Given a line segment with starting point as (0, 0) and ending point as (4, 4). Apply 30 degree rotation anticlockwise direction on the line segment and find out the new coordinates of the line.
b) Write Sutherland –Hodgeman polygon clipping algorithm. [5+5]

6.a) Summarize the octrees BSP Trees with examples.
b) Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0). Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object. [5+5]

OR

7.a) Describe the 3-D Viewing pipeline with neat sketch.
b) Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object. [5+5]

8.a) Discuss the Perspective Transformation with an example.
b) Explain the Depth-Buffer Method (Z-Buffer Method) with an example. [5+5]

OR

9.a) Describe the Binary Space Partitioning algorithm.
b) Explain the basic illumination models. [5+5]

10.a) List and discuss the general computer animation functions.
b) Illustrate the design of animation sequence. [5+5]

OR

11.a) Describe the raster animation with an example.
b) Explain the motion specifications with respect to animations. [5+5]

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